



Surgical anatomy of the adrenal glands

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INTRODUCTION

The adrenal glands are essential to normal physiologic functioning. Significant pathology requiring surgical intervention may result from hyperplasia, adenoma formation, or malignancy.

Adrenal surgery often is the primary treatment modality for a multitude of adrenal conditions. As a result, a strong working knowledge of adrenal embryology and anatomy is essential. Adrenalectomy can be performed by open or laparoscopic techniques with use of various approaches (anterior, lateral, or posterior) [1,2]. Regardless of the operative approach, the surgeon must have a complete understanding of the anatomy of the adrenal gland to avoid injury to vital adjacent structures and organs [3,4].

This topic will review the surgical anatomy of the adrenal glands. Surgical diseases of the adrenal gland, diagnosis, and treatment are discussed elsewhere. (See "[Clinical manifestations and diagnosis of adrenocortical carcinoma](#)" and "[Treatment of pheochromocytoma in adults](#)" and "[Evaluation and management of the adrenal incidentaloma](#)".)

FUNCTION

The adrenal glands are composed of two functionally distinct endocrine units, the adrenal cortex and medulla, contained within a single capsule ([figure 1](#)). Each has distinct embryologic, anatomic, histologic, and functional characteristics [5].

Adrenal cortex function — The adrenal cortex is divided into three functional zones ([figure 2](#)).

• The zona glomerulosa secretes mineralocorticoids (aldosterone), which regulate sodium and potassium homeostasis. (See "[Adrenal steroid biosynthesis](#)".)

• The zona fasciculata secretes glucocorticoids (most importantly, cortisol). (See "[Adrenal steroid biosynthesis](#)".)

• The zona reticularis secretes sex steroids (primarily androgens). (See "[Adrenal steroid biosynthesis](#)".)

The adrenal stem and progenitor cells reside near the adrenal capsule of the zona glomerulosa. Cells differentiate and undergo centripetal migration, moving from the periphery toward the medulla at approximately 13 to 20 micrometers per day under basal conditions. Stress can increase migration greatly. As cells migrate, they undergo lineage conversion and change their steroidogenic phenotype to match the functional zone they occupy [6].

Adrenal medulla function — The adrenal medulla synthesizes and secretes catecholamines, which modulate the body's sympathetic response to stress. The synthesis of catecholamines from the amino acid tyrosine is localized in the cells of the adrenal medulla and the organ of Zuckerkandl and is modulated by phenylethanolamine-N-methyltransferase (PNMT), an enzyme that converts L-norepinephrine to L-epinephrine. Because PNMT is limited exclusively to these cells, epinephrine-secreting tumors arise predominantly in the adrenal medulla and the organ of Zuckerkandl [7-9]. (See "[Clinical presentation and diagnosis of pheochromocytoma](#)".)

Abnormal function — Adrenocortical diseases are classified on the basis of whether there is hormone deficiency or excess. Disorders of adrenal hormone deficiency (eg, primary adrenal insufficiency, also known as Addison's disease) are not treated surgically. In contrast, many disorders of adrenal hormone excess do require surgical intervention, including:

- Benign cortisol, aldosterone, or less commonly, androgen-secreting adenomas (see "[Clinical manifestations and diagnosis of adrenocortical carcinoma](#)")
- Pheochromocytoma (see "[Treatment of pheochromocytoma in adults](#)" and "[Pheochromocytoma and paraganglioma in children](#)")
- Adrenal carcinoma (most secrete cortisol with or without androgens) (see "[Treatment of adrenocortical carcinoma](#)")
- A rare type of Cushing's syndrome due to bilateral micronodular hyperplasia (see "[Cushing syndrome due to primary pigmented nodular adrenocortical disease](#)")

The presence of a functioning autonomous adenoma in one gland may produce atrophy in the other gland via negative feedback mechanisms. As an example, in Cushing's syndrome, due to an adrenal adenoma, autonomous secretion of cortisol by the adenoma suppresses both corticotropin-releasing hormone (CRF) from the hypothalamus and adrenocorticotrophic hormone (ACTH) from the pituitary, leading to a decrease in steroid synthesis and eventual atrophy of the remaining adrenal gland. Recovery of an intact hypothalamic-pituitary-adrenal (HPA) axis and a normal-sized remaining adrenal gland may take up to two years after unilateral adrenalectomy [4,10-12]. Similar principles apply to chronic glucocorticoid therapy. (See "[Approach to continuing systemic glucocorticoid therapy in adults and children](#)", section on '[Overview of glucocorticoid-induced adrenal insufficiency](#)'.)

EMBRYOLOGY

The embryological development of the adrenal glands gives insight into the formation of accessory or aberrant glands and the widespread distribution of chromaffin tissue in sites away from the adrenal medulla, which is

important in the management of [pheochromocytoma](#) and [paraganglioma](#) [13]. Both cortical and medullary elements appear between the fifth and sixth weeks of fetal growth.

- **Development of the adrenal cortex** – The cortex is derived from mesothelial cells lying in the dorsal wall of the primitive coelom adjacent to the mesonephric tubules and gonadal ridges [14]. By the eighth week of fetal development, these cortical elements have differentiated into a thin outer definitive cortex and a thick inner fetal cortex. The fetal cortex actively produces fetal steroids during gestation but involutes rapidly after birth. Adrenocortical rests occur in up to 50 percent of newborn infants but tend to atrophy and disappear in the early postpartum period [3,4].

The development of adrenal zones occurs slowly after birth, in parallel with regression of the fetal cortex, and is not completed until late in the first year of life [7]. The definitive cortex persists and develops into the functional adrenal cortex, with distinct zonae glomerulosa and fasciculata present at birth. The zona reticularis develops during the first year of life.

- **Development of the adrenal medulla** – The adrenal medulla and sympathetic nervous system develop in concert. The medullary elements, the sympathogonia, migrate forward from both sides of the neurogenic crest to the paraaortic and paravertebral regions and along the adrenal vein toward the medial aspect of the developing adrenal fetal cortex.

Most extra-adrenal chromaffin cells regress. However, some cells remain and form the organ of Zuckerkandl, which is located to the left of the aortic bifurcation near the origin of the inferior mesenteric artery.

Extra-adrenal gland anatomy — True accessory adrenal glands, which contain both cortical and medullary tissue, are rare. Most extra-adrenal glands contain either cortical or medullary tissue.

- **Ectopic cortical tissue** – Ectopic cortical tissue often occurs in the areolar tissue adjacent to the kidney or in the pelvis. This ectopic tissue is usually found in relation to the sympathetic plexus and along the path of migration of structures arising from the urogenital ridge: epididymis, vas deferens, ovarian pedicle, broad ligament of the uterus, or within the ovary or testis ([figure 3](#)). Adrenocortical tissue also may be found in locations that are not explained by normal migration patterns of fetal tissues [14,15].
- **Ectopic medullary tissue** – Extra-adrenal chromaffin tissue may persist at any location along the migration path of neural crest cells, along the abdominal aorta, in association with the para-aortic sympathetic chain, the retroperitoneal celiac plexus, and the urinary bladder [14,16].
- **Extra-adrenal pheochromocytomas and paragangliomas** – Extra-adrenal chromaffin tumors are termed extra-adrenal pheochromocytomas and are also known as paragangliomas [17]. Extra-adrenal pheochromocytomas constitute 15 percent of adult and 30 percent of pediatric pheochromocytomas [17,18]. They are situated most commonly in the organ of Zuckerkandl, which is the collection of paraganglia located anterolaterally to the distal abdominal aorta between the origin of the inferior mesenteric artery and the aortic bifurcation. The second most common location of extra-adrenal pheochromocytomas is at the left renal hilum. Extra-adrenal pheochromocytomas can be multiple and have also been reported in the neck, posterior chest, atrium, and bladder [8,16]. (See "[Clinical presentation and diagnosis of pheochromocytoma](#)".)

SIZE AND LOCATION

The adrenal gland is approximately one-third the size of the kidney at birth. However, in the adult, the adrenal gland is only one-thirtieth the size of the kidney. This change in proportional size is a reflection of renal growth as well as the involution of the fetal adrenal cortex after birth.

At the onset of puberty, the gland is at its adult size and only increases slightly in weight over the course of adult life, with the exception of stress and pregnancy or the development of pathology [5]. A normal adrenal gland in an adult weighs approximately 4 to 6 grams. The left adrenal gland is larger and flatter than the right adrenal gland. The weight of each adrenal gland may increase by nearly 50 percent during times of stress and pregnancy. Pathologic glands may reach 700 grams [19].

The adrenal glands are retroperitoneal and are located on the superior medial aspect of the upper pole of each kidney ([figure 4](#)). Gerota's fascia and pararenal fat separate the adrenals from the ribs; the pleural reflection; and the subcostal, sacrospinalis, and latissimus dorsi muscles [14]. The adrenal glands often are not visible on direct inspection of the retroperitoneum, and their identification requires careful dissection and mobilization of adjacent structures and surrounding fat.

The normal adrenal cortex and medulla have different characteristics:

- The normal adrenal cortex is dark yellow and has a firm consistency and finely granular surface, allowing it to be differentiated from surrounding adipose tissue [10]. The cortex composes 80 to 90 percent of the volume of a normal gland.
- The central medulla is red-brown and is enclosed completely by the adrenal cortex, except at the hilum. The medulla constitutes 10 to 20 percent of the volume of a normal gland.

Right adrenal gland location — The right adrenal gland is pyramid shaped ("witch's hat") and lies above the upper pole of the right kidney, between the liver and the diaphragm.

Superiorly, the right adrenal gland abuts the bare area of the liver. Based on autopsy studies, nearly 10 percent of people develop hepatic-adrenal fusion as a result of loss of the fibrous tissue between the liver parenchyma and the cranial portion of the right adrenal gland [20]. Less frequently, the adrenals are adherent to the kidney.

The right adrenal gland has the following relationships to surrounding structures:

- The ventral-lateral area of the right adrenal is overlapped by the peritoneum between the liver, kidney, and hepatic flexure of the colon.
- The ventral-medial region is behind the inferior vena cava, separating the gland from the epiploic foramen anteriorly and the third portion of the duodenum and pancreatic head posteriorly. The body of the pancreas separates the right adrenal gland from the lesser sac and stomach. The thin medial border is related to the right celiac ganglion and the right inferior phrenic artery.
- The right adrenal gland may reside partially within the upper portion of the right paracolic gutter if the inferior layer of the right triangular or coronary ligament is particularly high.

Left adrenal gland location — The left adrenal gland is found between the kidney and aorta, near the tail of the pancreas and the splenic artery. The left adrenal gland has the following relationships to surrounding structures:

- The superior half of the anterior surface of the left adrenal gland is covered anteriorly by peritoneum of the lesser sac. This separates the gland from the cardia of the stomach and the posterior pole of the spleen. The avascular segment of the gastrocolic ligament commonly is divided during surgery on the left adrenal gland.
- There is no peritoneum inferiorly, where the gland contacts the pancreas and splenic artery. The transverse mesocolon attaches along the inferior border of the pancreas and is retracted inferiorly and medially for operative exposure.
- The medial border is adjacent to the inferomedial left celiac ganglion and to the left inferior phrenic and left gastric arteries, which ascend on the left diaphragmatic crus.
- The ventral aspect of the left adrenal gland is attached to the dorsal viscera of the stomach and to the medial border of the spleen and the body of the pancreas [3,14].
- Both the splenic vein and artery are inferior to the left adrenal gland.
- The left adrenal gland is located anterior to the origin of the celiac trunk and is separated from the aorta by several millimeters.

BLOOD SUPPLY

The blood flow to the normal adrenal gland is approximately 10 mL/minute. The blood supply to both the cortex and medulla increases during periods of stress. Adrenocorticotrophic hormone (ACTH) produces an immediate increase in blood flow to the adrenals.

Arterial and venous capillaries within the adrenal gland integrate the function of the cortex and medulla. Cortisol-rich effluent flows from the cortex to the medulla, where it stimulates the synthesis and activity of the enzyme phenylethanolamine N-methyltransferase (PNMT), leading to the conversion of norepinephrine to [epinephrine](#). Extra-adrenal chromaffin tissues lack this mechanism, thereby secreting mostly norepinephrine [8,14].

The adrenal glands have a rich arterial supply from three main groups of inflow vessels ([figure 5](#) and [figure 6](#)) [10,15]:

- **Superior suprarenal arteries** – The superior suprarenal arteries are derived from the inferior phrenic arteries, which pass just superior and medial to the adrenal glands. Each inferior phrenic artery gives off a series of branches to the ipsilateral adrenal gland before it supplies the diaphragm.
- **Middle suprarenal artery** – The middle suprarenal artery is derived from the aorta.
- **Inferior suprarenal renal arteries** – The inferior suprarenal arteries are derived from the adjacent renal artery.

Other adjacent vessels also may supply branches to the adrenal gland; these include the intercostal arteries, the left ovarian artery, and the left internal spermatic arteries. Because any arteries approaching the gland can branch and re-branch, the number of adjacent vessels entering it may be quite numerous. There is no constant position in which these arteries enter the gland. This anatomic detail is important because encountering these small vessels in the perirenal fat indicates proximity of the gland, and the feeding vessels will bleed unless cauterized or ligated [19].

VENOUS DRAINAGE

With rare exceptions, each adrenal gland has a single draining vein. Vascular control of the adrenal vein by the surgeon is easier on the left because the left adrenal vein is much longer than the right [10]. On the right, the adrenal vein is less than 1 cm in length and empties directly into the posterior inferior vena cava, creating a risk for injury and hemorrhage during surgery.

Venous drainage of the left adrenal gland — The left adrenal vein emerges at the hilum of the gland and is approximately 2 to 3 cm in length. The length of the left adrenal vein permits ready vascular control during left adrenalectomy [10,19].

The left adrenal (suprarenal) vein drains into either the left renal vein or the left inferior phrenic vein. The left adrenal vein passes inferomedially from the lower pole of the gland, receives the inferior phrenic vein, and follows an oblique downward course to enter the left renal vein. Occasionally, the left adrenal vein drains into the left inferior phrenic vein, which then empties into the left renal vein or crosses over the aorta to enter directly into the inferior vena cava.

Venous drainage of the right adrenal gland — The right adrenal (suprarenal) vein emerges from the hilum and enters the posterior segment of the inferior vena cava at a 45° angle. The right adrenal vein often cannot be exposed until the gland is circumferentially mobilized, because the vein is typically less than 1 cm in length.

The origin of the right adrenal vein may be obscured by an enlarged gland or tumor. Additional smaller veins are found in 5 to 10 percent of right adrenal glands. Rarely, aberrant veins may drain into the right hepatic vein or right renal vein. In addition, small direct hepatic branches draining from the posterior aspect of the liver into the vena cava may join the adrenal vein and can be torn during adrenalectomy [10,19].

LYMPHATIC DRAINAGE

The adrenal gland has two lymphatic plexuses: one in the medulla and one deep to the adrenal capsule.

Most of the adrenal lymphatics terminate in the lateral aortic lymph nodes and the paraaortic nodes near the diaphragmatic crus and origin of the renal artery. Hence, when operating on a suspected malignant adrenal tumor, adjacent paraaortic and paracaval nodes must be evaluated for evidence of local metastases [3,10].

The remaining lymphatics traverse the diaphragm and drain toward the thoracic duct or the posterior mediastinum, thus explaining the pattern of metastatic spread of malignant tumors.

INNERVATION

Innervation of the adrenal glands is via visceral afferent fibers arising from the celiac, aorticorenal, and renal autonomic ganglia in the retroperitoneum ([figure 7](#)). These fibers connect with the posterior vagus nerve, phrenic nerve, and greater and lesser splanchnic nerves ([figure 8](#)).

The nerve fibers provide sensory or indirect vasomotor innervation as they traverse the adrenal cortex and terminate in the medulla as preganglionic sympathetic fibers. The adrenal medulla is a postsynaptic sympathetic nerve and functions as a neuroendocrine transducer [8].

SUMMARY

- **Embryology** – The embryological development of the adrenal glands gives insight into the formation of accessory or aberrant glands and the widespread distribution of chromaffin tissue in sites away from the adrenal medulla along the paraortic chain, which is important in the consideration of pheochromocytoma and paraganglioma. (See '[Embryology](#)' above.)
- **Size and location** – The adrenal glands are retroperitoneal and are located on the superior medial aspect of the upper pole of each kidney. (See '[Size and location](#)' above.)
- **Blood supply** – The adrenal glands have a rich, multiple arterial supply from three main groups of vessels: the superior suprarenal arteries, the middle suprarenal artery, and the inferior suprarenal renal arteries. (See '[Blood supply](#)' above.)
- **Venous drainage** – Vascular control of the adrenal vein is more difficult on the right than the left because the right adrenal vein is less than a centimeter in length and empties directly into the posterior inferior vena cava, which creates a risk for injury and hemorrhage. (See '[Venous drainage](#)' above.)

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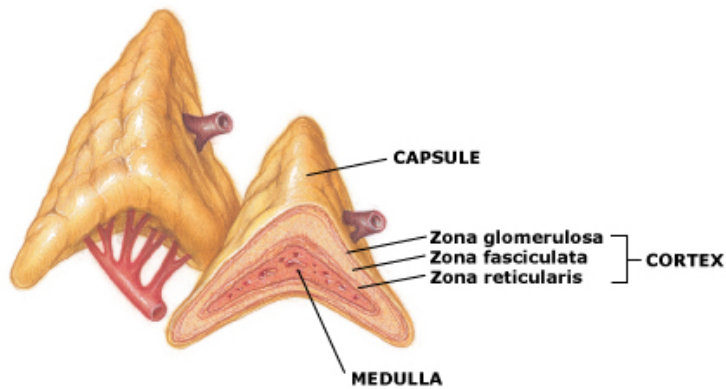
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GRAPHICS

Figure 1: Cross-sectional anatomy of the adrenal gland

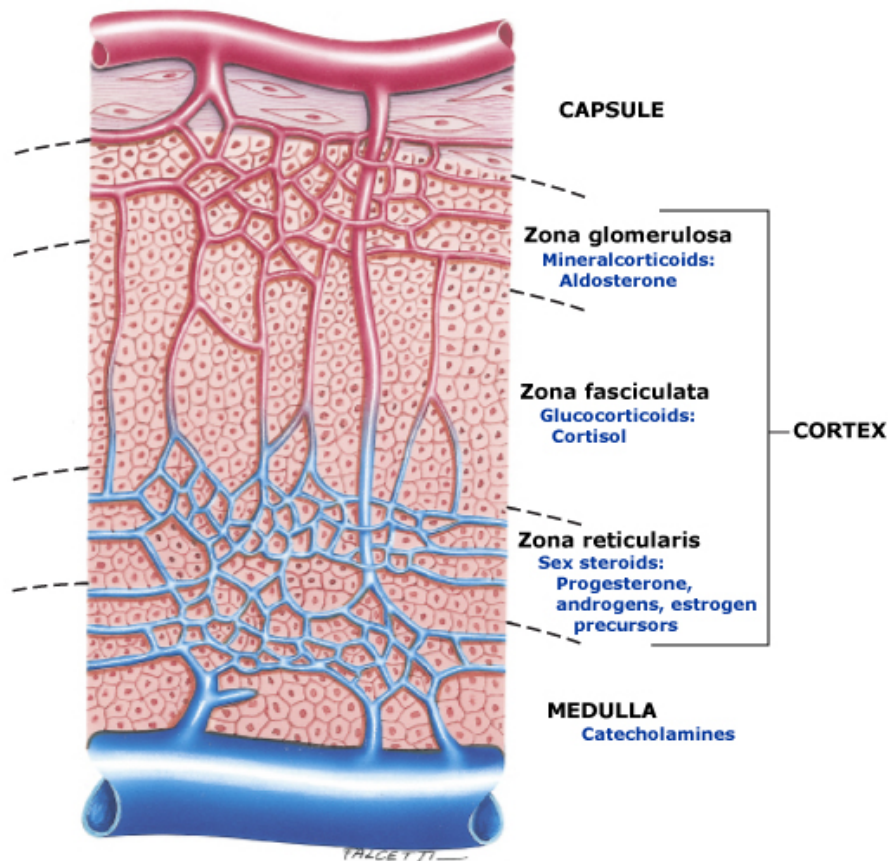


Adrenal glands showing cortex and medulla.

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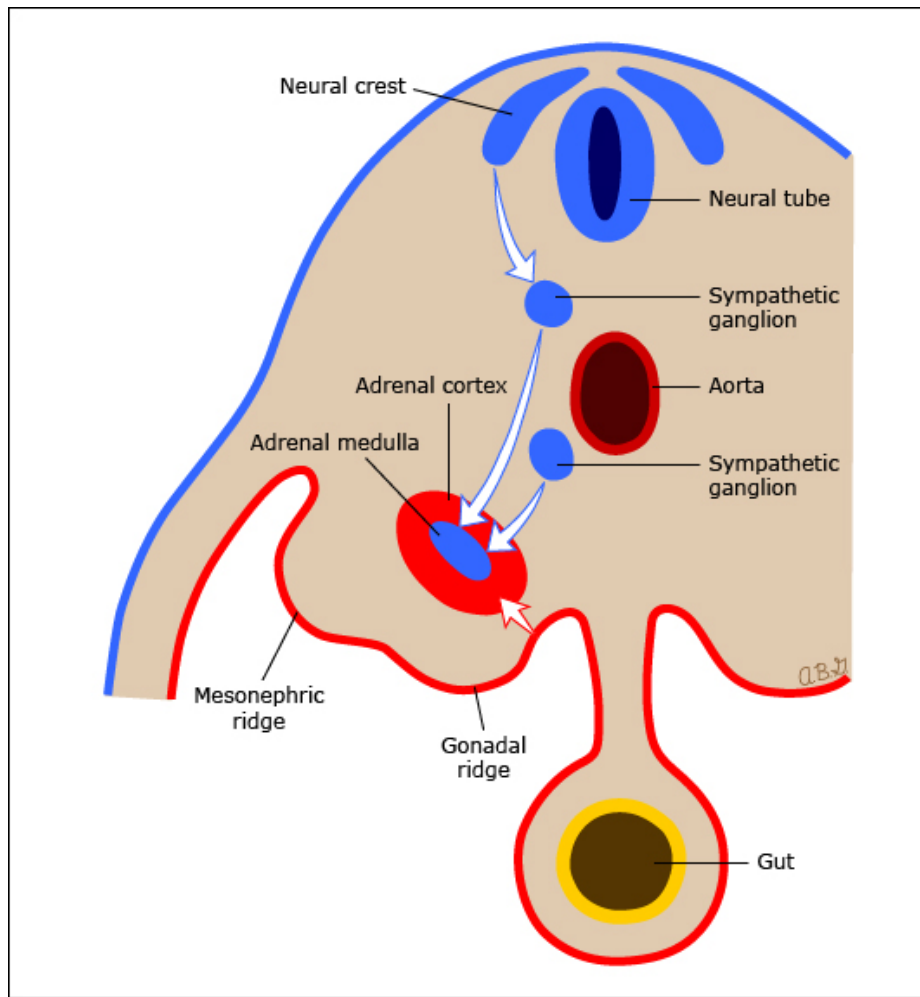
Figure 2: Zones of adrenal tissue



Schematic drawing of the left adrenal gland showing the microstructure of the gland on a transverse cut. Note the capsular circulation and the connections through the sinusoids to the medullary vein.

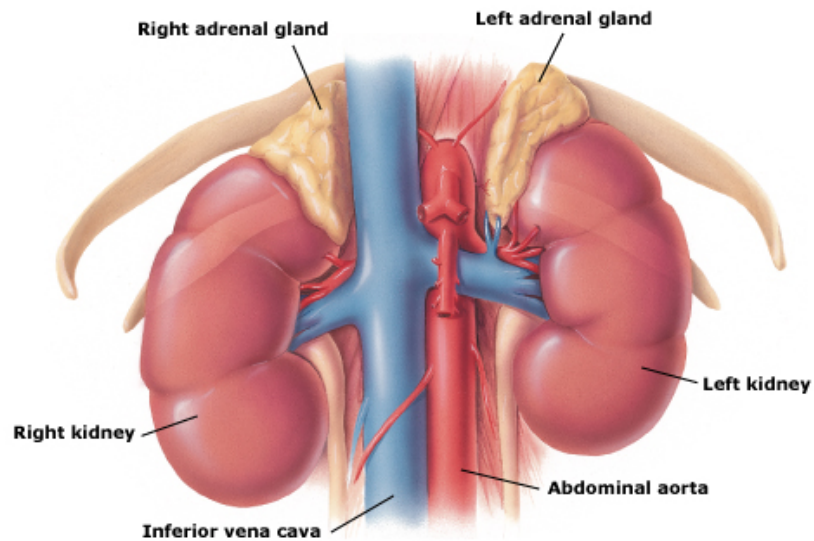
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Figure 3: Development of the adrenogenital primordium

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Figure 4: Adrenal glands in relation to the kidneys and great vessels

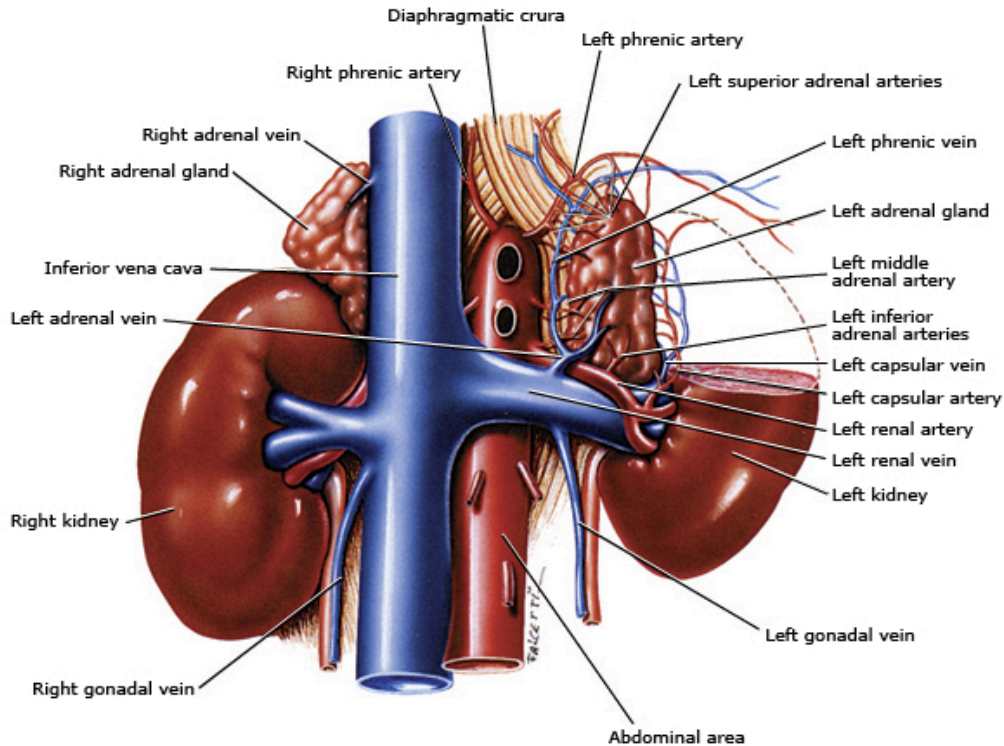


The adrenal glands are retroperitoneal and are located on the superior medial aspect of the upper pole of each kidney. This illustrates the frontal or intra-abdominal view of the kidneys and adrenal glands.

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Figure 5: Adrenal blood supply

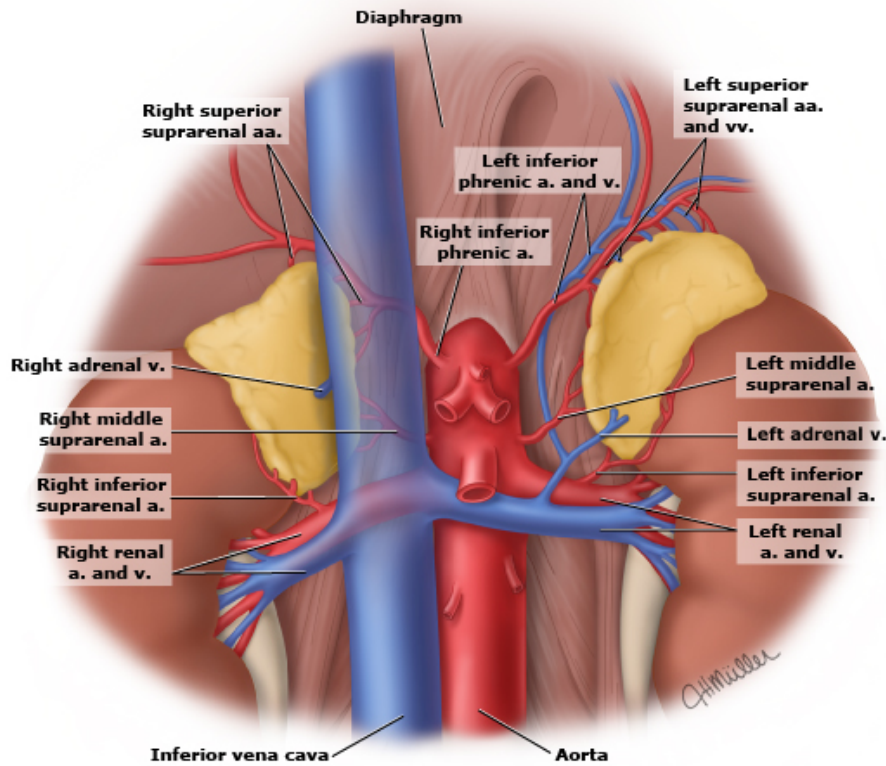


Schematic drawing showing the inferior vena cava, the aorta, the renal veins and arteries, and the relationship of these structures with the adrenal glands. Note the arterial and venous supply of the left adrenal gland. Only the vein of the right adrenal gland is shown.

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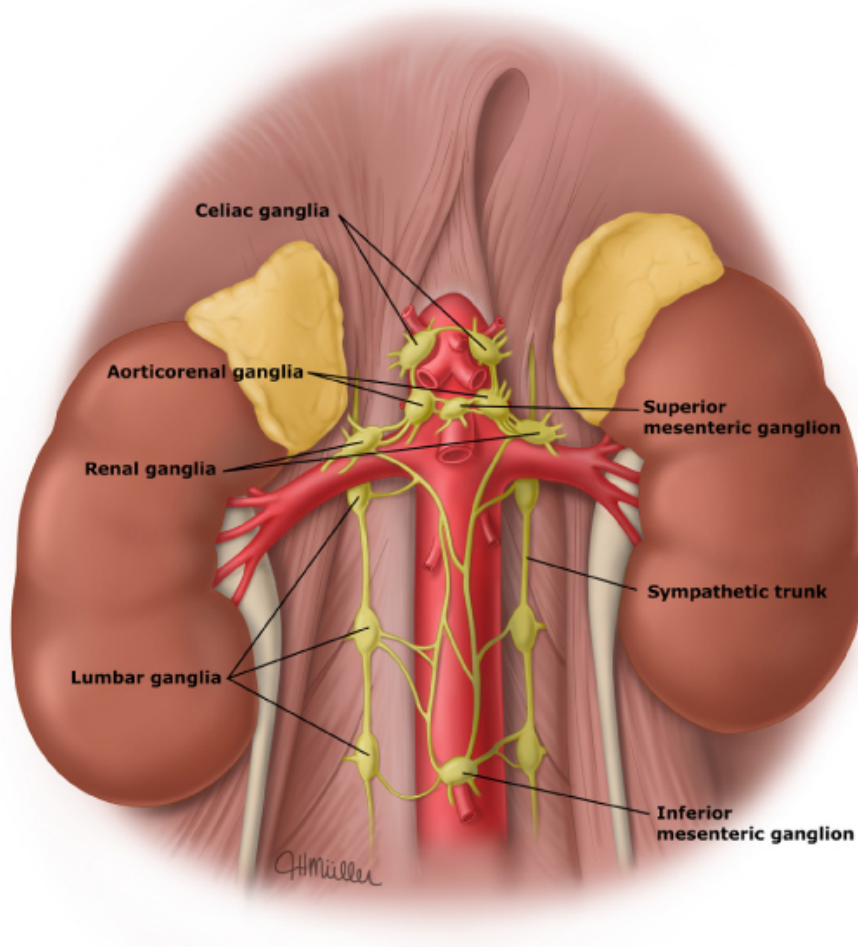
Figure 6: Blood supply of the adrenal glands



Three main groups of vessels supply the adrenal gland: the superior suprarenal arteries, derived from the inferior phrenic arteries; the middle suprarenal artery, derived from the aorta; and the inferior suprarenal renal arteries, derived from the adjacent renal artery.

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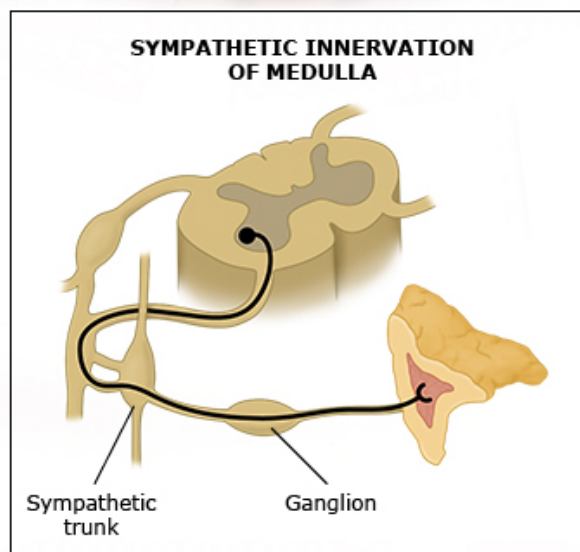
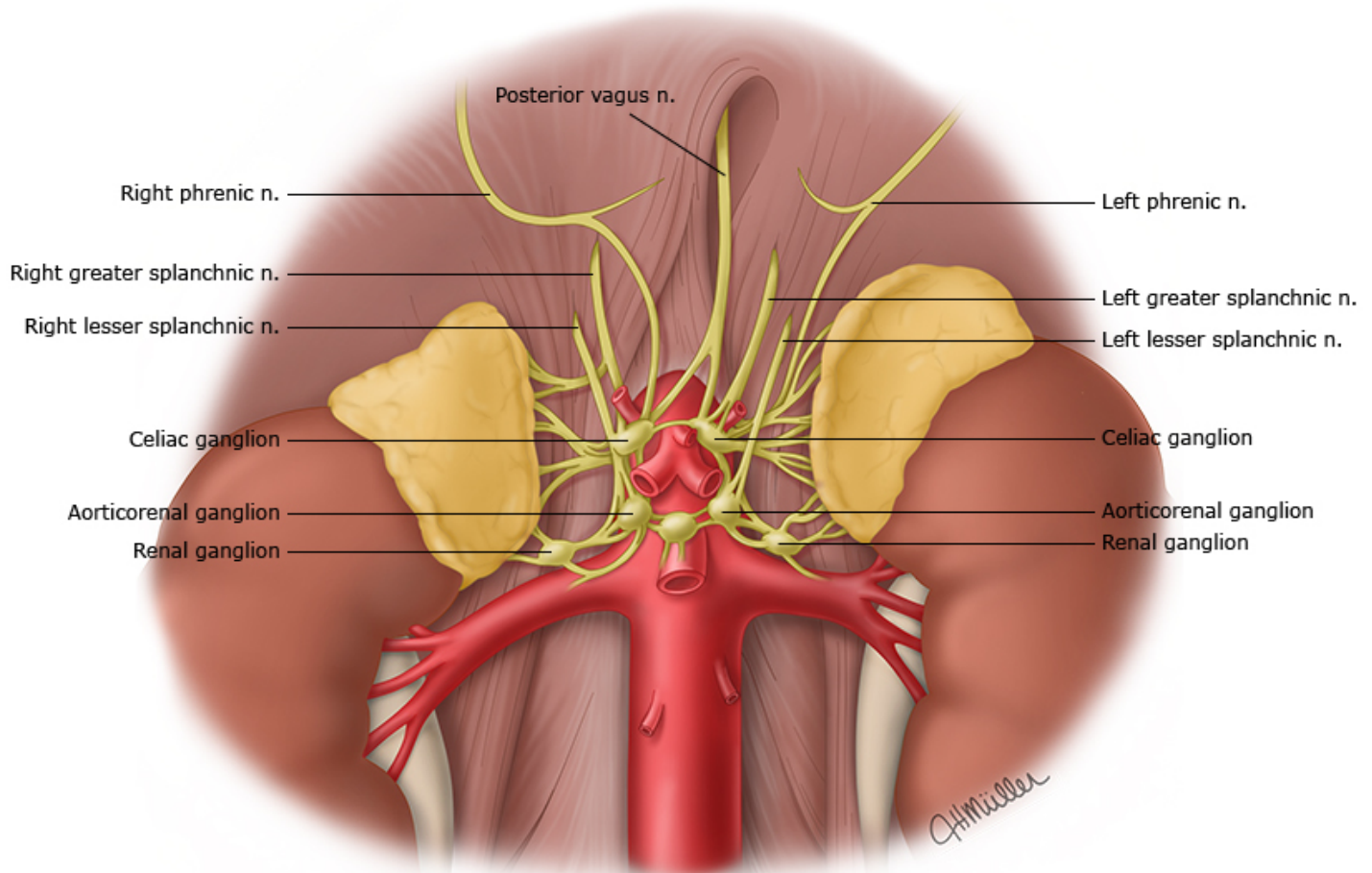
Figure 7: The autonomic ganglia in the retroperitoneum



Innervation of the adrenal glands is via visceral afferent fibers arising from the celiac, aorticorenal, and renal ganglia.

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Figure 8: Innervation of the adrenal glands



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The innervation of the adrenal glands is via visceral afferent fibers arising from the celiac, aorticorenal, and renal ganglia. These fibers connect with the posterior vagus nerve, phrenic nerve, and greater and lesser splanchnic nerves.

The nerve fibers provide sensory or indirect vasomotor innervation as they traverse the adrenal cortex and terminate in the medulla as preganglionic sympathetic fibers.

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Sanziana Roman, MD, FACS No relevant financial relationship(s) with ineligible companies to disclose. **Leslie Wu, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Sally E Carty, MD, FACS** Other Financial Interest: Jaypee Brothers [Endocrine surgery]. All of the relevant financial relationships listed have been mitigated. **Wenliang Chen, MD, PhD** No relevant financial relationship(s) with ineligible companies to disclose.

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